

Summary: New Directions for Bent Crystals (CERN Courier)

Overview

This article from the *CERN Courier* (the news magazine for the high-energy physics community) serves as a high-level report on a major recent milestone at the LHC: the successful preliminary demonstration of **double-channelling**. It highlights how bent crystals are moving beyond basic beam cleaning and opening "new frontiers in particle physics."

The Core Experiment

The graphic and text outline a groundbreaking setup where **2 TeV LHC protons** are steered not just by one, but by **two consecutive bent crystals**.

The setup involves:

1. **The First Bent Crystal:** Intercepts the proton beam and provides the initial channelling and deflection.
2. **The Second Bent Crystal:** Positioned downstream to catch the already-deflected beam and channel it a second time.

Measurement and Tracking

To prove that particles successfully navigated both crystals, the researchers used advanced tracking detectors:

- **Fibre Tracker Measurement Data:** Used to track the trajectory of the particles between or after the crystals.
- **Pixel Tracker Measurement Data:** Provides high-resolution hit maps to precisely identify where the particles ended up.

The Results

The trackers allowed the scientists to clearly distinguish between two types of particle fates:

- **Single-channelled:** Protons that were successfully deflected by the first crystal but missed or dechannelled in the second.
- **Double-channelled:** The ultimate goal—protons that successfully entered the Lindhard acceptance angle of the first crystal, survived the bend, and then successfully entered and survived the bend of the second crystal.

The Big Picture

While the previous academic paper (TWOCRIST) detailed the rigorous testing of the crystals themselves, this *CERN Courier* piece celebrates the actual operational success of the double-crystal setup in the LHC. Successfully double-channelling 2 TeV protons is the exact proof-of-concept needed for future exotic experiments (like ALADDIN), which plan to place targets between two crystals to measure the spin precession of short-lived baryons.

